isonavi

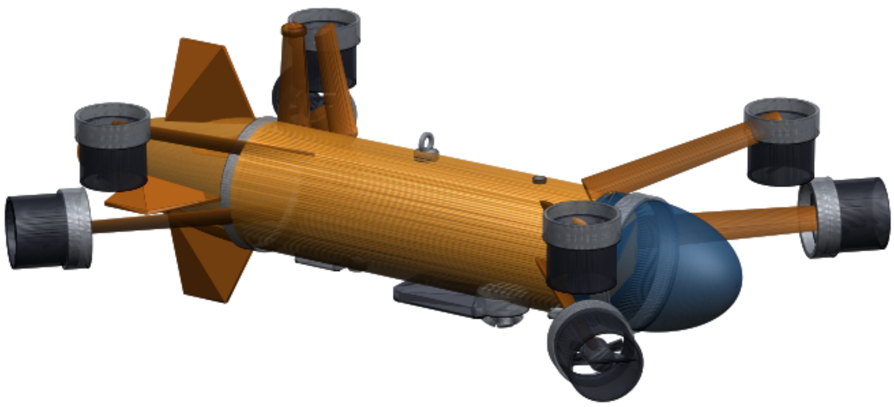
Autonomous Underwater Reconnaissance and Assessment for Post-Flood Disaster Response

Technical Design and Verification Report

Simulation, autonomy and acoustic perception for zero-visibility, high-current Indian river conditions

Summary. Post-flood search and bridge foundation inspection in India depend on divers who cannot safely work in monsoon flood conditions: the water is opaque with suspended sediment, the current exceeds human capability, and submerged structural debris makes entry unsafe. This report presents a complete, verified simulation and autonomy stack for an underwater vehicle designed for exactly those conditions, together with quantitative results from an end to end autonomous mission over a modelled post-collapse river site.

Four results are established. First, the commercial platform normally assumed for this role cannot perform it: an open frame vehicle of that class holds station only to $0.96\,\mathrm{m\,s^{-1}}$ against a design current of $2.4\,\mathrm{m\,s^{-1}}$. Second, a purpose designed faired vehicle raises that envelope to $2.69\,\mathrm{m\,s^{-1}}$ on drag derived from its own geometry, and completes a 573 m autonomous survey with a mean navigation error of 0.322 m across 5 seeds, 0.15 m in the run reported in detail, without any external positioning. Third, submerged vehicles are located from the resulting bathymetric map with no training data at all, at a mean localisation error of 1.80 m, reducing an unsearchable survey box to roughly 24.4 georeferenced contacts to dive. Finally, the complete mission was executed with the autonomy running on a RISC-V flight computer, reproducing the simulation result to within centimetres inside the real-time control budget, which moves the maturity claim from simulated to hardware-validated.



Station keeping envelope	$2.69\,\mathrm{m\,s^{-1}}$	Navigation error, no external fix	0.322 m mean of 5
Autonomous mission	573 m, 14.9 min	Bathymetric map error	0.285 m
On a RISC-V flight computer	0.107 m	Endurance, slack to design current	17.8 to 1.2 h
Depth rating	50 m	Parts cost, one airframe	27.6 lakh INR

Every number in this report is generated from the measured result files by a script in the repository that accompanies it

`github.com/AUV-VITP/isonavi`

Contents

1 Problem and operational context

1.1 The gap

India experiences recurrent monsoon flooding that destroys road and rail crossings and sweeps vehicles into rivers. Two capabilities are needed immediately afterwards, and neither is currently available at acceptable risk:

1. **Search.** Locating submerged vehicles and casualties in turbid, fast moving water.
2. **Structural assessment.** Measuring scour around bridge foundations to decide whether remaining spans are safe.

Both are performed today by divers. In flood conditions this is close to impossible. The reference scenario used throughout this report is the collapse of the Savitri river bridge on NH-66 at Mahad, Maharashtra, on 2 August 2016, when a British era masonry arch bridge failed during monsoon flood and carried two state transport buses and several private vehicles into the river. Recovery took days, for precisely these reasons.

Optical imaging is not a candidate. At the suspended sediment concentrations of a monsoon flood, modelled here at $3.2\,\mathrm{g\,L^{-1}}$, visibility is effectively zero. The system is therefore built around acoustic perception throughout.

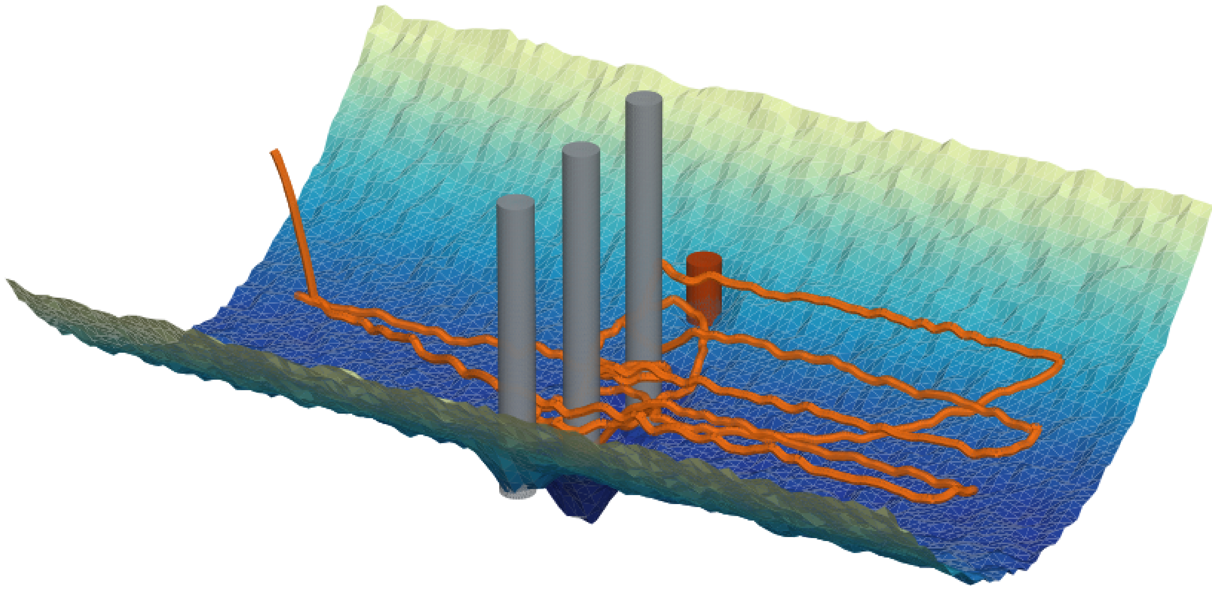


Figure 1: The survey reach the vehicle was designed for, at a vertical exaggeration of 2.6. Three intact piers and one collapsed pier, shown in orange, stand on the modelled bathymetry, and the scour hollows around the pier bases are part of that model rather than an annotation. The orange track is the trajectory actually flown in the reference mission, logged from the simulator, not a sketch of a survey pattern.

1.2 Statutory driver

Beyond emergency response, IRC:SP:35 requires periodic underwater inspection of bridge foundations, and the Dam Safety Act 2021 mandates inspection of submerged structures. Scour depth is the quantity that determines whether a pier is at risk. Measuring it currently requires divers or reservoir drawdown. In the system described here it falls out of a routine survey automatically, which converts an occasional, hazardous, qualitative inspection into a repeatable quantitative measurement.

2 Why the standard platform does not work

Proposals in this space normally assume an open frame commercial ROV of the BlueROV2 class. The first engineering result of this work is that such a vehicle cannot perform the mission, and the reason is not control tuning but installed thrust against hydrodynamic drag.

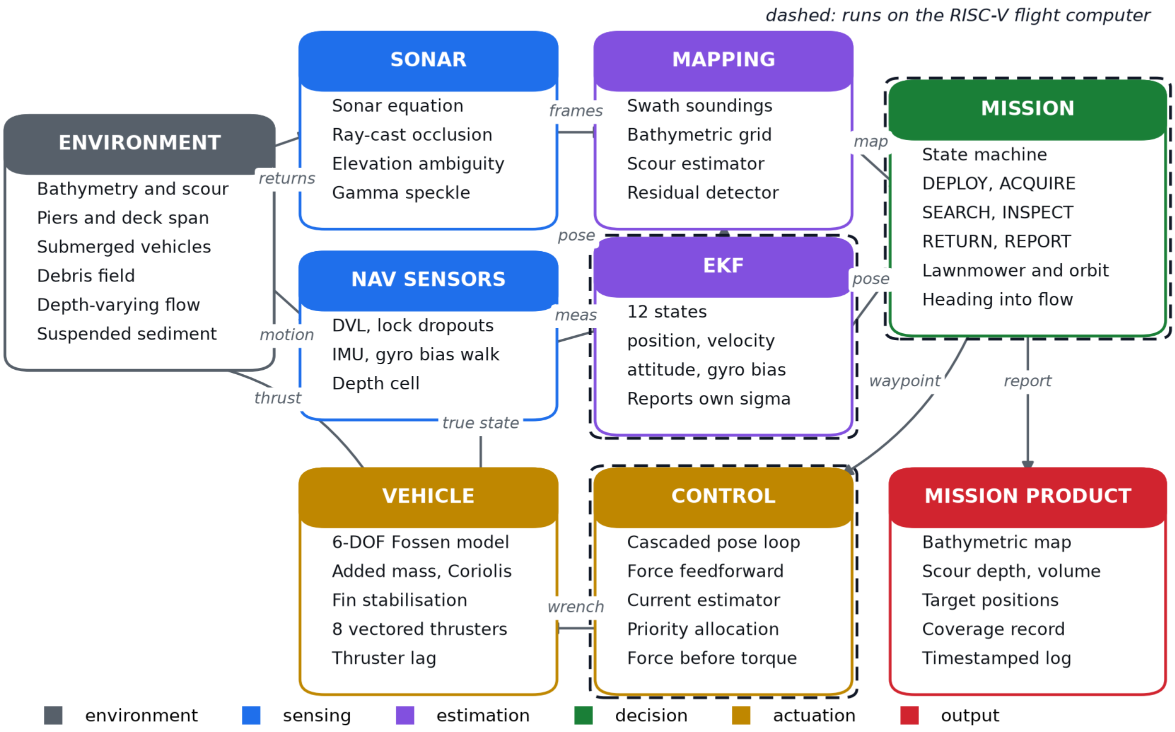


Figure 2: System architecture. The vehicle acts only on estimated quantities; ground truth is used solely to score the result afterwards.

For steady motion the surge drag is $X_u v + X_{u|u|} v|v|$. Setting this equal to the largest pure surge wrench the thruster layout can produce before any single thruster saturates gives the maximum current the vehicle can hold station against. For the open frame baseline this is $0.96\,\mathrm{m\,s^{-1}}$. The modelled site runs at $2.4\,\mathrm{m\,s^{-1}}$ at the surface. The vehicle is swept away, and no controller can prevent it.

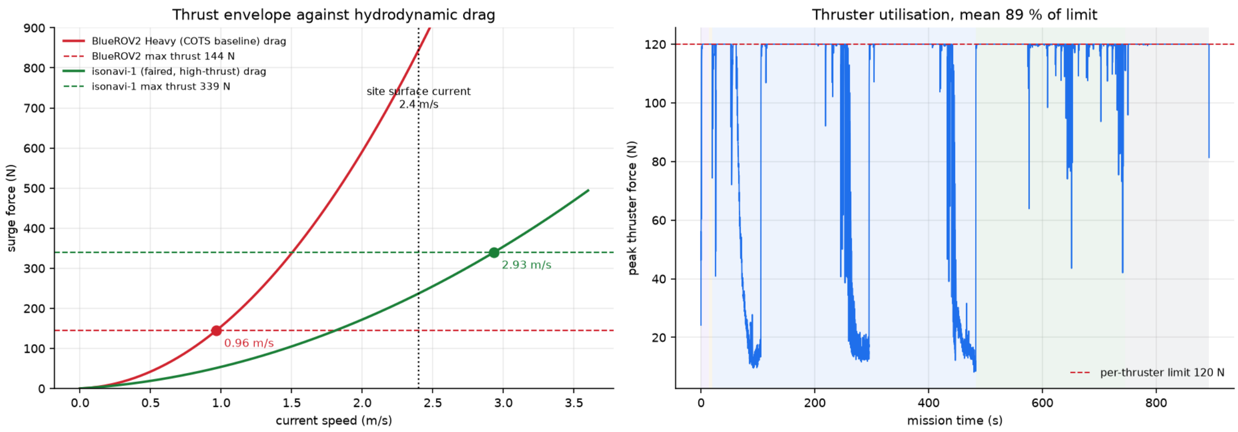


Figure 3: Left: surge drag against current speed for both vehicles, with the saturated thrust limit of each. The intersection is the maximum holdable current. The open frame baseline is out of envelope at the design condition by a factor of more than two. Both curves use the drag the dynamics model carries; Section 2.2.3 rederives it from the built geometry and moves this vehicle's envelope to $2.69\,\mathrm{m\,s^{-1}}$. Right: thruster utilisation over the reference mission for the purpose designed vehicle.

2.1 The design response

Two changes recover the envelope, and a third is forced by the first:

- **Faired hull.** Axial quadratic drag is reduced by roughly a factor of four relative to an open frame. Drag is deliberately left high broadside, which also gives useful passive weathervaning into the flow.

- **Larger thrusters and longer moment arms.** Force and torque share one thrust budget, so moment arms are sized to buy attitude authority without consuming the surge thrust that holds the vehicle in the current.
- **Fixed stabilising fins.** A faired hull is directionally unstable. The added mass asymmetry between axial and transverse directions produces a destabilising Munk moment proportional to $(Z_{\dot{w}} - X_{\dot{u}})\, u_r w_r$ whenever the hull moves at an angle of attack. Buoyancy restoring is far too weak to counter it. Fins are therefore a requirement, not a refinement, and they only stabilise when the vehicle flies nose first.

Table 1: Vehicle comparison at the design condition.

	Open frame baseline	**isonavi-1**
Mass	13.5 kg	28.0 kg
Axial quadratic drag $X_{u\|u\|}$	141	32
Thrust per unit	51 N	120 N
Stabilising fins	none	yes
Maximum holdable current	0.96 m s^{-1}	2.93 m s^{-1}
Station keeping at 1.96 m s^{-1}	swept, 14.0 m error	holds, 0.40 m RMSE

2.2 Vehicle general arrangement

The vehicle is defined parametrically from the same numbers the simulation uses, so the drawing is the vehicle that was simulated rather than an illustration of one. The hull is a Myring profile, the standard low drag axisymmetric body for autonomous underwater vehicles: an elliptical nose, a parallel mid-body, and a fined tail. It is 1074 mm long on a 180 mm diameter, a slenderness of 6.0.

The faired hull is itself the pressure boundary, as on survey vehicles of this class, rather than a free flooding fairing wrapped around a separate cylindrical housing. Two things forced that choice. A separate housing leaves the buoyancy to syntactic foam, and the foam volume required here does not fit in the annulus between a 150 mm housing and the skin. Trimming that arrangement to a useful stability margin then needs ballast in a bulb below the hull, and a keel bulb is a snag hazard in exactly the debris field this vehicle exists to survey. With the hull as the pressure vessel the ballast sits inside on the keel line and nothing protrudes below the skin except the acoustics.

2.2.1 The mass and stability budget closes

Every component the vehicle carries is listed with a real mass at a real station, and three quantities are then solved rather than chosen: the hull size, so displaced volume equals the value the simulation uses for buoyancy; the trim ballast mass, so dry mass equals the simulated mass; and the ballast station, so the vehicle floats level.

Table 2: Mass budget by group. The solve closes on all three constraints simultaneously.

Group	**Mass (kg)**	**Share**
Trim ballast	9.03	32.3 %
Structure	8.73	31.2 %
Power	4.00	14.3 %
Propulsion	2.72	9.7 %
Sensors	2.22	7.9 %
Avionics	1.30	4.6 %
Total	28.0	
Displaced volume 0.0282 m^3, net buoyancy 1.96 N positive		
Longitudinal trim 0.000 mm, hull volume closes to 0.001 %		

Because those three are solved, everything else the layout produces is a prediction rather than a decoration. Net buoyancy comes out 1.96 N positive, which is the right sign: the vehicle surfaces on a power failure rather than staying on the bed. The 6.3 kg of trim ballast is also the payload growth margin, since at fixed displacement it can be traded for instrumentation without touching the hull.

The layout also carries the hardware that makes a vehicle operable rather than merely buildable: two sacrificial anodes, a lifting eye sited over the centre of gravity so the vehicle leaves the water level, a mast

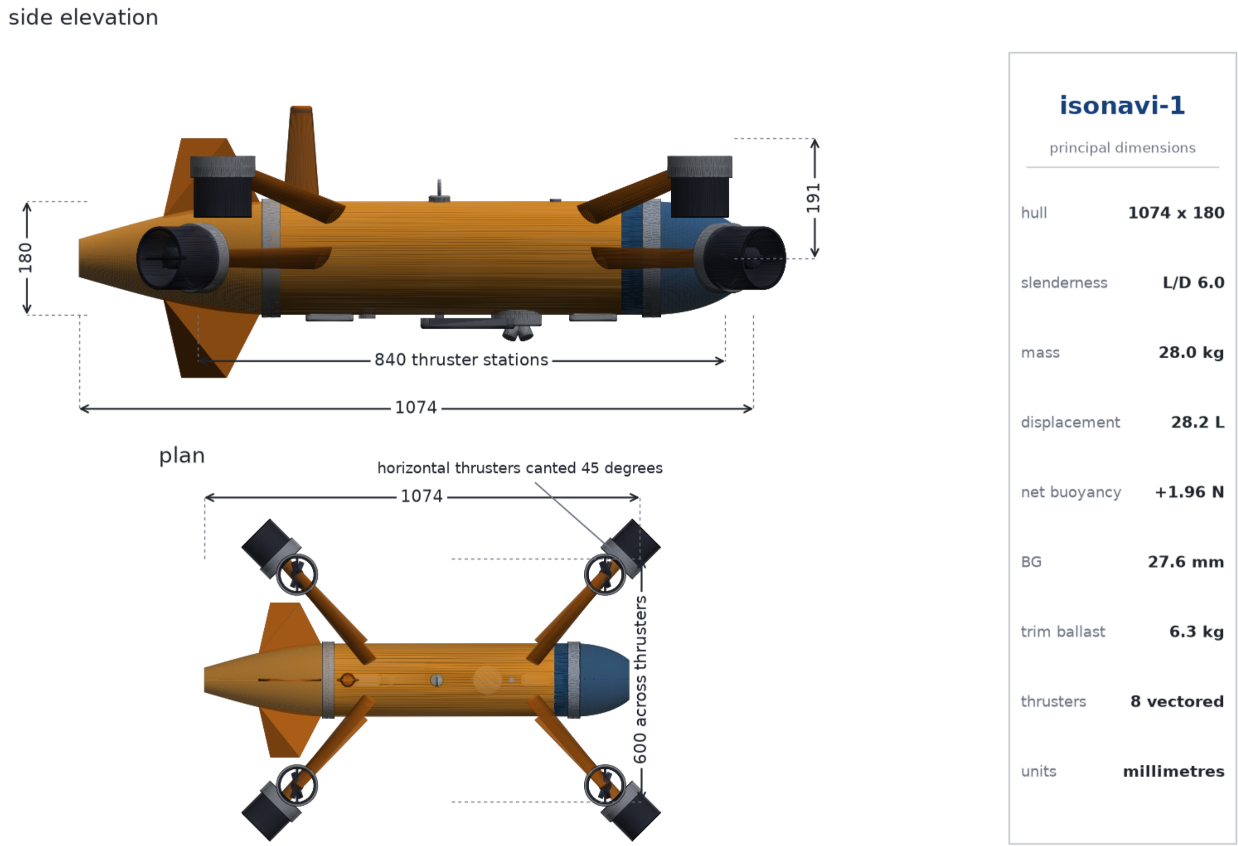


Figure 4: Principal dimensions. The hull form, the thruster stations and the 45° cant all come from the parameters the dynamics model uses.

carrying GPS, Iridium and a strobe, a bulkhead of wet mateable connectors for the eight thruster runs, and clamp band joints so the bay can be opened. Each has a mass and a station, so each is in the budget rather than drawn on afterwards.

Two kilogrammes of the trim is carried outside on a burn wire release. Dropping it takes the vehicle from 1.96 N positive to 19.85 N positive, a tenfold increase, which is what recovers it from entanglement or a flat battery. A vehicle that is only just positive will sit on the bed when its thrusters stop; one that can shed ballast will not.

2.2.2 The hull has to survive the depth

Making the faired hull the pressure boundary is only defensible if it holds, so it was sized. For a thin shell under external pressure the governing failure mode is not strength but elastic instability: the skin buckles long before it is crushed. At the 50 m design depth the membrane hoop stress is 10.8 MPa against a laminate strength of 250 MPa, a factor of 23.2, so strength is not the question.

Stability is. An unstiffened 4 mm skin of this diameter collapses at 66 m, which would leave almost no margin on a 50 m rating. Ring frames shorten the unsupported length and the critical pressure rises steeply as they do. At the 200 mm spacing modelled, collapse moves to 482 m, a factor of 7.3, giving 9.6 on the design depth.

Table 3: Pressure hull at 50 m design depth. The frames are load bearing, not tidiness: they are what makes a 4 mm skin viable.

Mode	Capacity	Factor
Membrane strength	250 MPa	23.2
Buckling, no frames	66 m depth	1.3
Buckling, framed	482 m depth	9.6

The thruster pylons were checked the same way. The horizontal pods stand furthest outboard and so carry the worst cantilever on the vehicle, 214 mm with one thruster at its 120 N saturation limit. Root bending

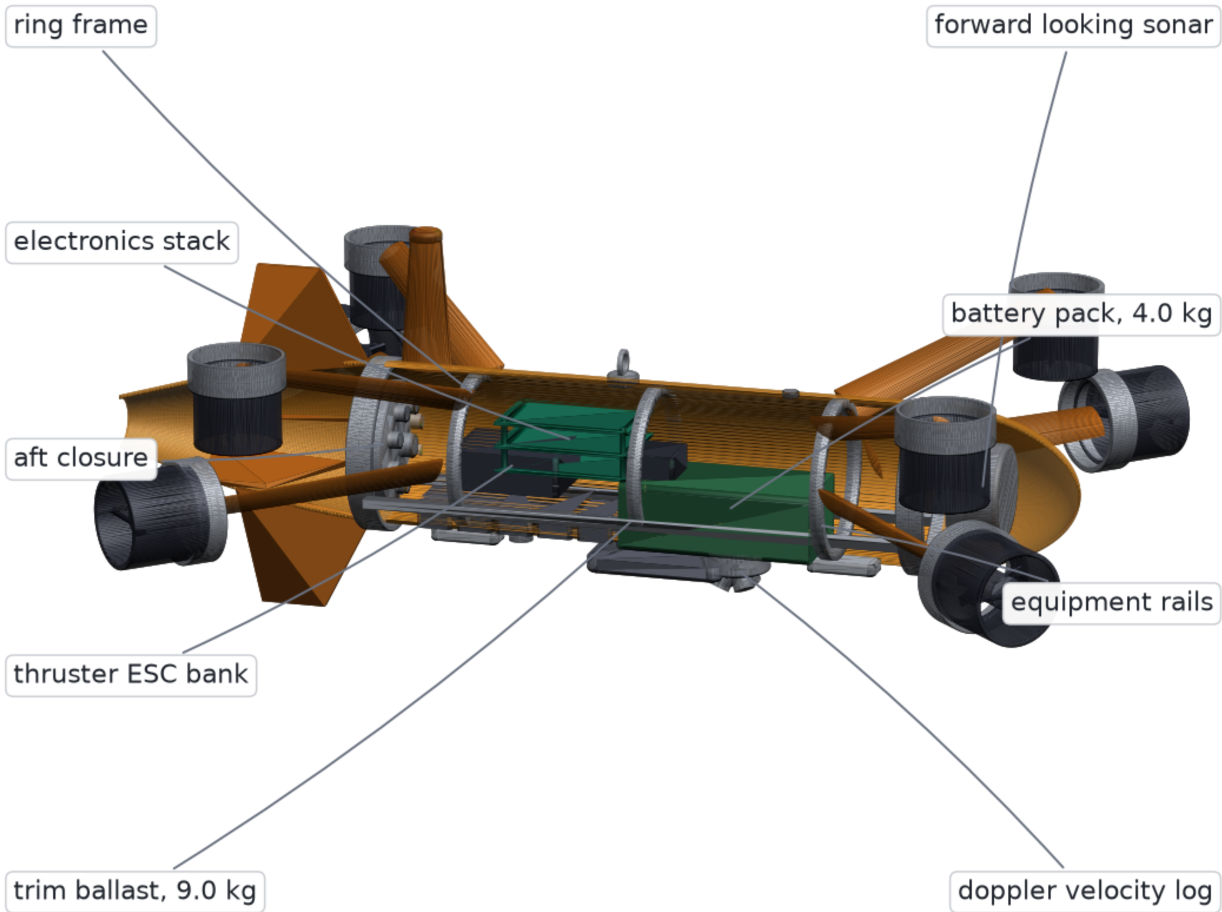


Figure 5: Internal arrangement on the centreline. Battery low and forward, electronics and speed controllers aft of it, trim ballast along the keel, sonar behind the acoustic window.

reaches 10.3 MPa, a factor of 24, so the pylons are sized by stiffness and by the flow rather than by strength.

These are closed form hand calculations of the kind used to size a shell before analysis. They are not a substitute for finite element work or a pressure test, and both belong in the build phase.

2.2.3 Where the drag actually comes from

The dynamics model has been carrying drag and added mass coefficients that were estimates made before the vehicle had a shape. It has one now, so they can be derived from it. Bare hull friction is taken from the ITTC 1957 correlation line with a form factor for a body of revolution, and the appendages are built up over the ducts, pylons and fins as modelled.

Table 4: Axial quadratic drag build up at $2\,\mathrm{m\,s^{-1}}$, in newtons per $\mathrm{m^2\,s^{-2}}$. The hull is not what makes this vehicle draggy.

Contribution	Coefficient	Share
Bare hull, friction and form	1.3	3 %
Four horizontal thruster ducts	21.2	55 %
Four vertical thruster ducts	14.7	38 %
Pylons and fins	1.7	4 %
Total from geometry	38.9	
Value used in simulation	32	ratio 1.21

Two things follow. The first is reassurance: the coefficient the simulation has been using is within 21 % of one derived from the geometry, which is close for a quantity that was previously an assertion.

The second is a design finding. The bare Myring hull contributes 3 % of axial drag. The eight thruster ducts contribute 93 %. Refining the hull form on this vehicle would buy almost nothing, because its drag is

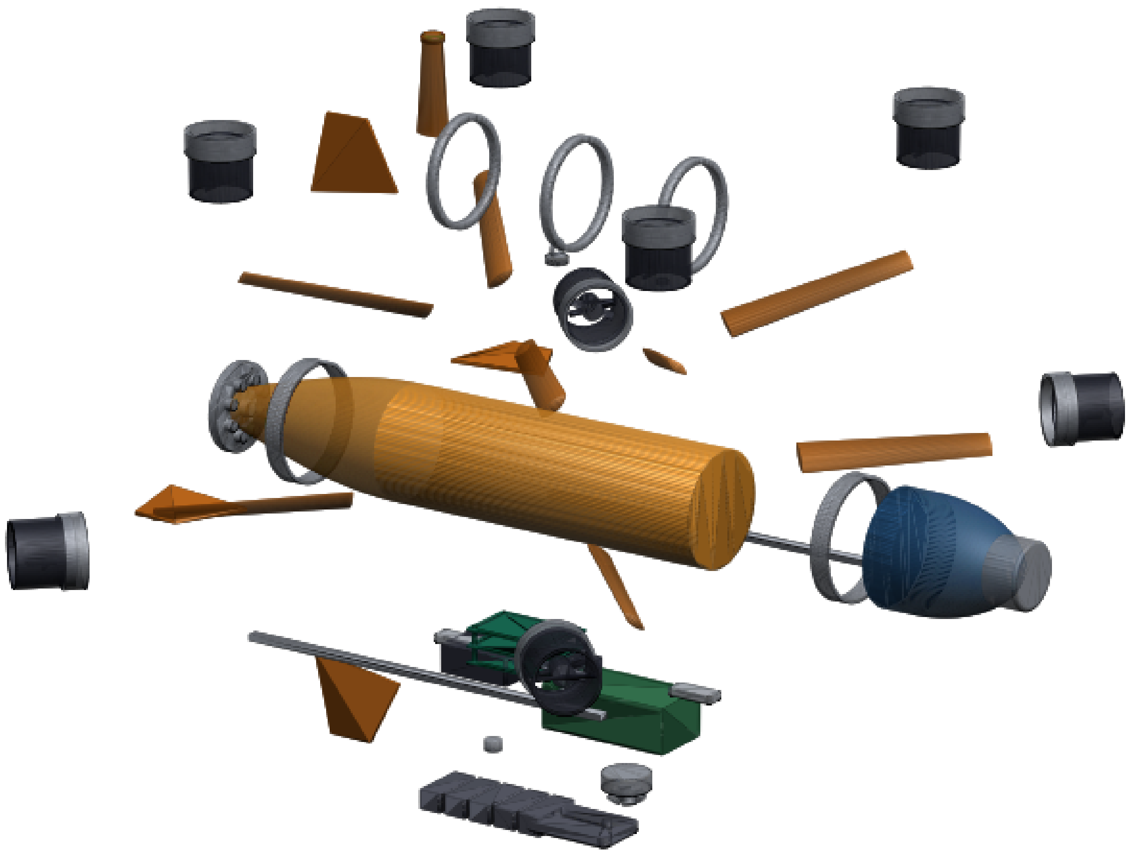


Figure 6: The same vehicle as a parts breakdown. Every line of the mass budget is a solid in the model, 52 parts in all: the eight thruster pods with their ducts and pylons, the ring frames, the nose dome, the fins, the acoustic window, the internal stack and the trim ballast. The budget in Table 2 is therefore a property of the geometry rather than a list kept alongside it, which is what allows the hull size, the ballast mass and the ballast station to be solved rather than chosen.

set by the hover authority bolted to it, not by its shape. If this vehicle is ever to go faster or further, the thrusters are what to fair or retract, and that is not a conclusion available without the CAD.

Added mass was estimated from Lamb's coefficients for the prolate spheroid of the same slenderness plus the water entrained by the ducts, giving 6.1 kg axially against 12 kg assumed, and 25.9 kg laterally against 42 kg. The simulation is the more sluggish of the two in both axes, which makes it the harder control problem and so the safer direction to have erred in.

The station keeping envelope was re-checked at the higher drag, because the entire case for this vehicle rests on it. It falls from $2.93\,\mathrm{m\,s^{-1}}$ to $2.69\,\mathrm{m\,s^{-1}}$ against a site surface current of $2.4\,\mathrm{m\,s^{-1}}$. The margin is therefore 1.12 rather than the 1.22 the simulation implies.

That the envelope survives being recomputed against the vehicle's own geometry, rather than against an assumed coefficient, is the result worth having: it is the check a design at this stage usually defers until a tow tank contradicts it. The margin is real without being generous, and it converts into a hard design bound. The appendage drag budget cannot grow by more than a quarter without buying more thrust. That bound is now a measured number rather than an assumption, and confirming it is exactly what the tow tank work in Section 12 is for.

2.2.4 What happens when a thruster is lost

A vehicle sent into a debris field will eventually lose a thruster, to line, rubble or flooding. Eight units driving six degrees of freedom leaves two spare, so the question is not whether the allocation is over-determined, which it plainly is, but whether the survivors can still produce the particular wrench the mission needs while holding attitude. That is a linear program, not an inspection of the matrix: for each failure, the largest force along an axis with zero net moment and every surviving unit inside its saturation limit.

Intact, the vehicle makes 339 N of surge and 480 N of heave, holding station to $2.69\,\mathrm{m\,s^{-1}}$. Station keeping at the site needs 277 N.

Table 5: Single thruster failures. Vertical failures cost heave and nothing else; horizontal failures halve surge.

Failure	Surge	Envelope	Holds site
None	339 N	$2.69\,\mathrm{m\,s^{-1}}$	yes
Any one vertical unit	339 N	$2.69\,\mathrm{m\,s^{-1}}$	yes
Any one horizontal unit	170 N	$1.83\,\mathrm{m\,s^{-1}}$	no

Losing a vertical thruster costs heave authority and nothing else. Losing a horizontal one costs half the surge, and the reason is geometric rather than incidental: the four horizontal units are canted in opposing pairs, so the single survivor on the light side has to balance the two on the heavy side, and the pair it balances against is capped by its own limit.

Heading offset does not recover it. Turning off the flow to present a better thrust direction also presents the hull broadside, and broadside quadratic damping on this hull is 210 against 38.9 in surge. Sweeping heading and carrying that penalty honestly, the best degraded envelope is $1.86\,\mathrm{m\,s^{-1}}$, against $1.83\,\mathrm{m\,s^{-1}}$ nose on. The fins cannot help either, because what binds is sway, not yaw.

The result is precise rather than binary, which is the useful kind. The vehicle is single fault tolerant for attitude and for depth. For station keeping at the design current it is not: after a horizontal failure it holds to $1.86\,\mathrm{m\,s^{-1}}$ and must otherwise abort downstream, which is always available because the flow assists it. Of the 28 possible double failures, 6 still hold station, and the worst pair, both units on one side, leaves no surge at all.

If single fault tolerance at the design current were made a requirement, the remedy is 196 N per thruster instead of 120 N, or six horizontal units instead of four. Both cost drag, mass and money. Single fault tolerance at the design current is therefore a purchase and not a redesign, and knowing its price to that precision is what lets the trade be made deliberately instead of discovered in the water.

2.2.5 What the budget caught

The separation between centre of buoyancy and centre of gravity sets the passive roll and pitch restoring moment. It is not a free parameter: it falls out of where the mass actually sits. The layout puts it at 27.6 mm. The dynamics model had assumed 85 mm.

85 mm is not reachable in a hull this size. It requires the centre of gravity to sit 15 mm off the bottom skin, which no arrangement of a battery and a pressure housing achieves. The figure was an estimate made before the vehicle had an internal layout, and it was optimistic by a factor of three.

The simulation was corrected to the CAD value, 27.3 mm, and the full mission and repeatability suite re-run. The layout has since gained its ancillary hardware and now gives 27.6 mm; the models are therefore one percent apart, which the sensitivity below shows to be immaterial. The results move very little, and the reason is worth stating because it is the same argument that justified the fins in the first place. At survey speed the restoring moment is dominated by the fins, which contribute $440\,\mathrm{N\,m\,rad^{-1}}$ at $2\,\mathrm{m\,s^{-1}}$ against $7.5\,\mathrm{N\,m\,rad^{-1}}$ from the buoyancy offset. Correcting a term worth two percent of the total changes the net stabilising moment at speed by four percent. At hover the buoyancy term is all there is, and it halves, but it remains positive and the vehicle remains upright.

The layout is then cross checked against the geometry kernel, which found a second error. The solve had been placing the hull's displaced volume at the hull mid-length, which a Myring form does not satisfy: the tail is nearly twice the length of the nose, so the volume centroid sits 41 mm aft of mid-length. The centre of buoyancy was therefore in the wrong place, and the trim ballast the solve returned was 105 mm out of position. The budget had been reporting level trim while describing a vehicle that would have floated nose up. Computing the centroid properly fixes it, and the analytic and kernel figures now agree to 0.01 mm.

This is the loop worth having. Both errors were real, both were in the direction of flattering the design, and in each case it was the act of building the vehicle properly that caught them.

2.3 Consequences for the mission plan

Because the hull is only stable flying nose first, and because a broadside turn at the design current is unrecoverable within the available sway thrust, two constraints propagate into the autonomy layer: every commanded heading points into the flow, and survey legs run along the flow axis rather than across it. This is an example of vehicle physics dictating mission structure, and it is captured directly in the planner.

3 Acoustic simulation

The core of this work is a physics based forward looking sonar simulator. It forms images from the active sonar equation applied per range and bearing cell, rather than texturing a depth buffer, and it runs on CPU alone.

3.1 Image formation

For a resolution cell at slant range r insonified by beam b,

$$\mathrm{EL}(r,b) = \mathrm{SL} - 2\,\mathrm{TL}(r) + \mathrm{BS}(r,b), \qquad \mathrm{TL}(r) = 20\log_{10} r + \alpha r, \tag{1}$$

with backscattering strength following a Lambert like angular law $\mathrm{BS} = \mu_{\mathrm{material}} + 10n\log_{10}(\cos i)$ plus a specular term for smooth hard faces. The absorption coefficient $\alpha = \alpha_{\mathrm{water}}(f) + \alpha_{\mathrm{sediment}}(C)$ uses the pure water term of the Francois and Garrison formulation, which at 750 kHz gives 61 dB km^{-1}. The sediment term represents excess attenuation from suspended solids in flood water and is an engineering approximation, stated as such.

Three properties of real imagery are reproduced explicitly, because they are what a perception model actually keys on:

1. **Acoustic shadow.** Ray casting resolves occlusion, so every object throws a shadow whose length encodes its height. Shadow geometry is the primary recognition cue in sonar imagery.
2. **Elevation ambiguity.** A forward looking sonar collapses the vertical aperture: all scatterers within the vertical beamwidth at a given slant range fold into one pixel. This is reproduced by casting many elevation rays per beam and summing them under the vertical beam pattern.
3. **Speckle.** Coherent imaging gives Rayleigh distributed amplitude, so intensity is Gamma distributed with shape equal to the number of looks.

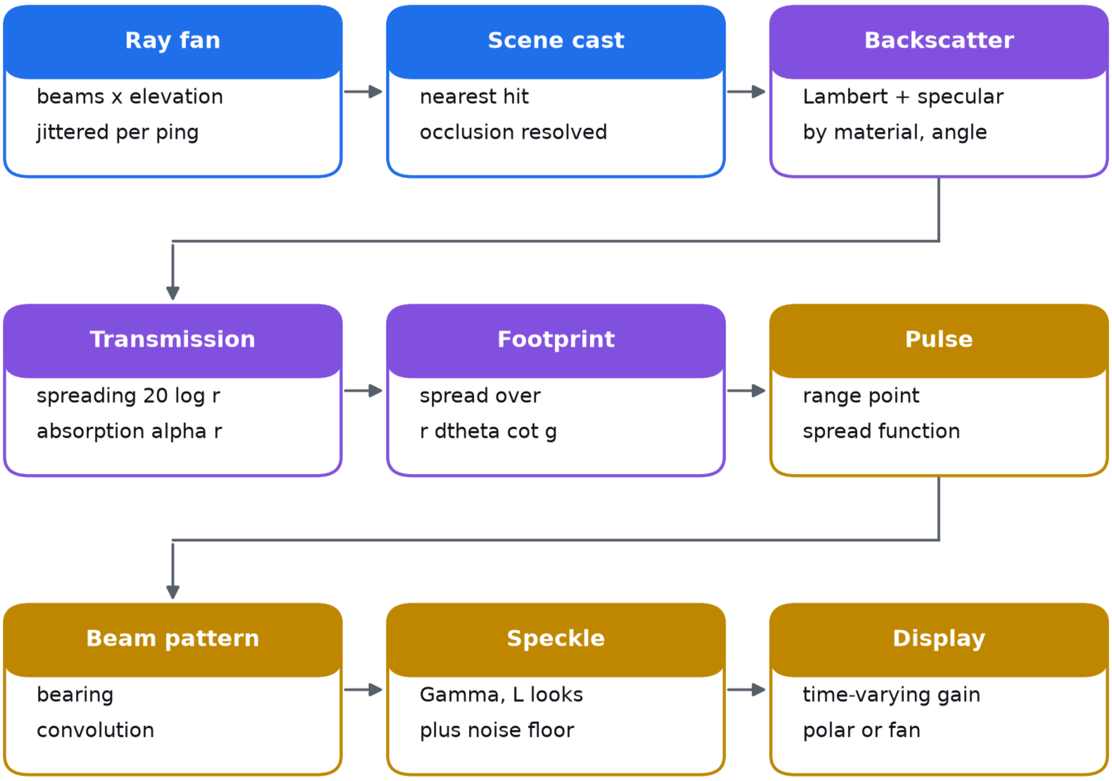


Figure 7: Sonar image formation. Every stage is a physical effect rather than a rendering convenience.

3.2 Footprint treatment

Each elevation ray represents a solid angle, not a point. Its intersection with a surface spans a finite range interval, and its energy is deposited across that interval. For an elevation beam of angular width $\Delta\theta$ striking a surface at grazing angle g, the along range extent is

$$L = r\,\Delta\theta\,\cot g, \qquad \sin g = |\hat{d}\cdot\hat{n}|. \tag{2}$$

Deriving L from local geometry rather than from neighbouring rays matters at object edges. There the adjacent ray may strike a surface tens of metres further away, and a midpoint rule would smear that ray's energy across the very shadow the object is casting. This was observed directly during verification and corrected.

3.3 Verification

The model is verified against physical expectations rather than by inspection. Ray primitive intersections are checked against analytic values to 10^{-9}. For a steel target on a silt bed the model produces a 12.7 dB target return and a −43.6 dB acoustic shadow behind it, with the seabed return falling smoothly with range. One ping costs roughly 150 ms on eight CPU cores, which is faster than the 2 Hz the vehicle actually uses.

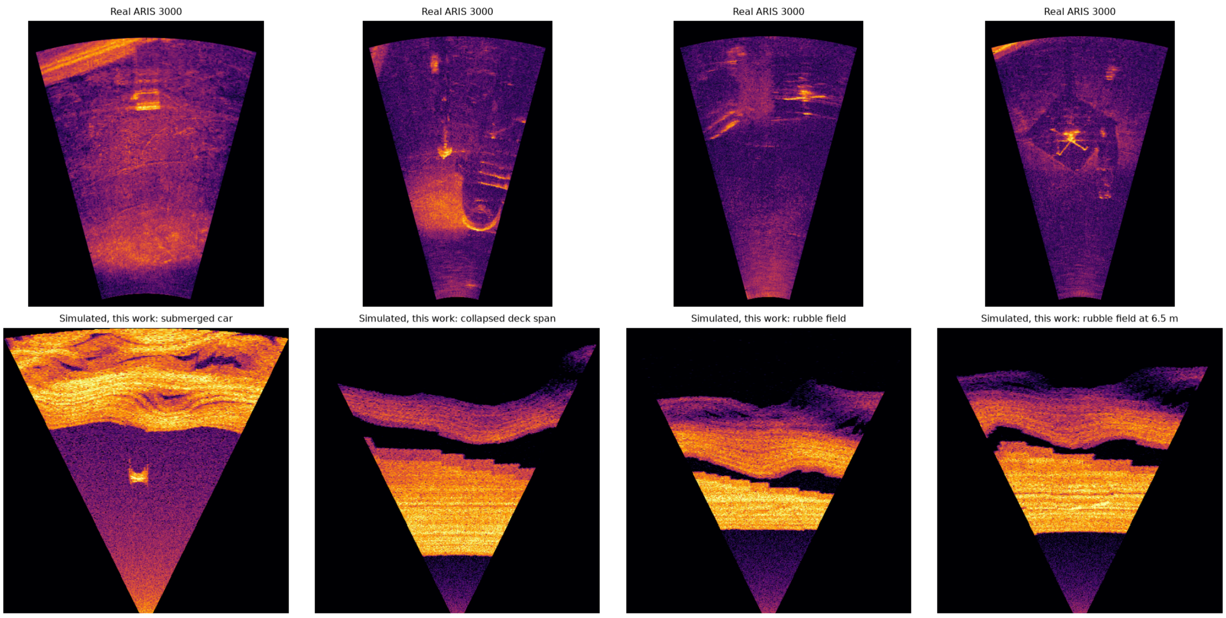


Figure 8: Top: real ARIS Explorer 3000 captures from the public marine debris dataset. Bottom: frames from this simulator, posed at the same short range, near bottom geometry the real captures were taken at, so the bed return fills the fan in both rows. The first simulated panel is the submerged car: a bright return with the acoustic shadow it casts behind it, which is the primary recognition cue in sonar imagery. The remaining panels are bed relief, where the shadowed far sides of slopes produce the same effect. Both rows share speckle texture and brightness that falls with range. The submerged bus is not shown: it lies on the lip of the channel, so every approach sees it edge on with no bed behind it to shadow.

4 Autonomy stack

4.1 Navigation

There is no GPS underwater. The pose estimate comes from a twelve state extended Kalman filter carrying position, body velocity, attitude and gyro bias, corrected by a Doppler velocity log, a depth cell and the inertial attitude solution. Sensor models include the failure behaviour that actually limits the solution, in particular loss of DVL bottom lock over soft sediment, which the research literature identifies as a principal failure mode.

Horizontal position is never directly observed. It is dead reckoned from DVL velocity, so its error grows without bound. The filter therefore carries and reports its own position covariance, which the mission layer can use to decide when a position reset against a mapped structure is required.

4.2 Control

A cascaded controller maps position error to a desired ground velocity, then to a body wrench. The current is fed forward as a *force*, being the thrust needed to balance hull drag at the estimated flow speed. Adding it to the velocity setpoint instead would be incorrect, because the inner loop regulates ground referenced velocity from the DVL, and inflating that setpoint commands the vehicle to fly downstream rather than hold position. This error was made and corrected during development, and it is worth stating because it is not obvious.

The ambient current is estimated without any flow sensor, by inverting the drag model against the applied thrust and the measured ground velocity. Over the reference mission this recovers the true current to within $0.01\,\mathrm{m\,s^{-1}}$ in steady conditions.

Thrust allocation is prioritised. When demand exceeds what the thrusters can deliver, translational force is preserved and attitude torque is scaled back, because losing station in fast water is unrecoverable whereas a slow attitude response is not. Uniform scaling of the whole wrench does the opposite: during development a large yaw demand was found to collapse delivered surge thrust from 340 N to 67 N, which alone caused the vehicle to be swept away.

5 Mapping, scour and target detection

Every ping yields returns across the swath, which are projected through the estimated vehicle pose and accumulated into a gridded bathymetric map. Because the map is built in the estimated frame, navigation drift propagates into it, so all mapping results below measure the whole chain rather than the sonar alone.

5.1 Scour quantification

The undisturbed bed level around a pier is taken as the median of an annulus outside the scour hole. Depth is the deepest excursion below that datum and volume is the integral of the depression. If the pier surroundings are not sufficiently sounded the result is reported as unmeasured rather than as a spuriously shallow value.

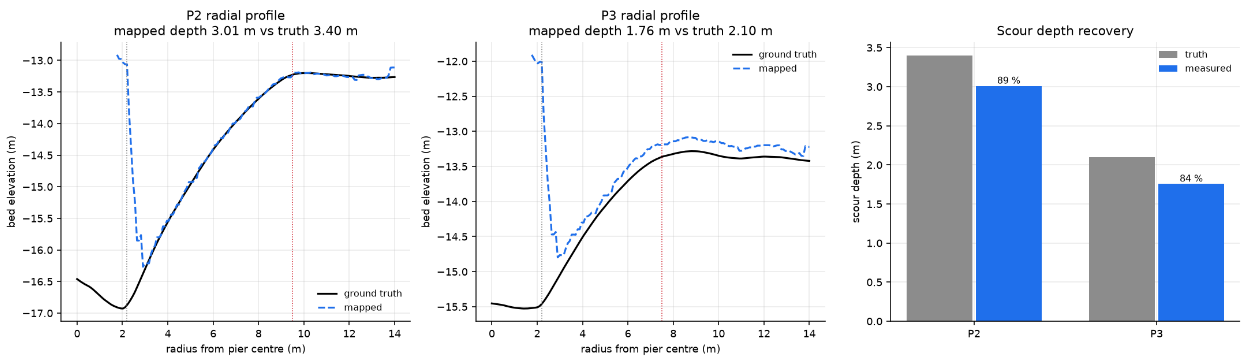


Figure 9: Radial bed profiles around the inspected piers, mapped against ground truth, and the resulting scour depth recovery.

5.2 Detection without training data

The bathymetric residual above a smooth estimate of the bare bed isolates objects standing proud of the riverbed. The bed is estimated by grey scale morphological opening, which removes compact objects while following broad bathymetry including scour hollows. The structuring element must exceed the largest object of interest: a window shorter than an 11 m bus absorbs the bus into the bed estimate and hides it entirely.

This gives a purely geometric detector that needs no labelled sonar at all. That matters because no public training data exists for submerged Indian buses, and it is the target class that matters most.

6 Learned acoustic perception

A YOLOv8s detector was trained on the public marine debris forward looking sonar dataset, captured with an ARIS Explorer 3000 and comprising 2035 real images across 11 classes. On the held out test split it reaches 99.0 % mAP at IoU 0.5, 97.4 % precision and 98.2 % recall.

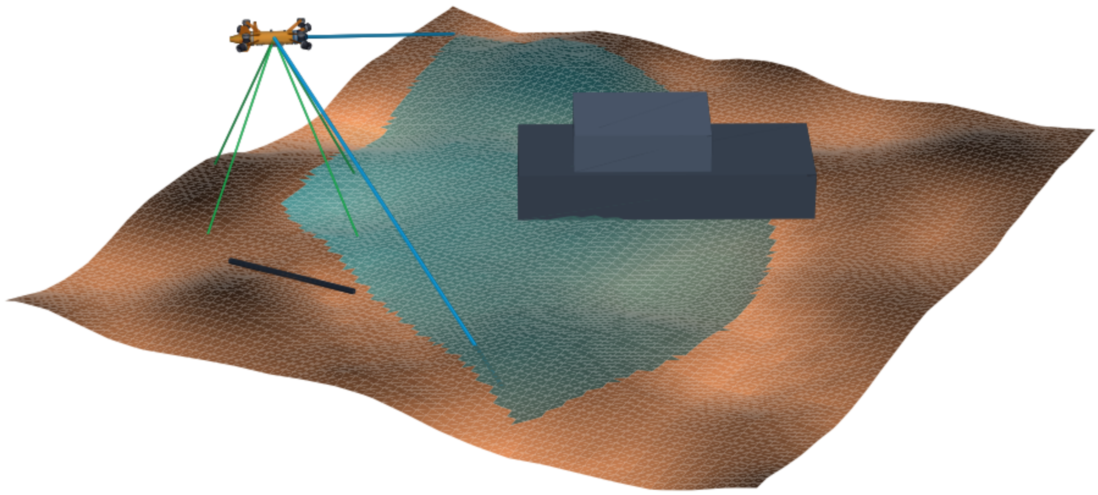


Figure 10: A detection pass over a submerged car, with the vehicle drawn at true scale against it. Cyan is the area of bed lying inside the 130° sonar aperture at this instant, bounded by the two blue aperture edges. Green are the four Doppler velocity log beams at the 30° Janus angle that holds the navigation solution together with no external fix. The dark bar is 2 m. Every element is the geometry the simulator actually uses.

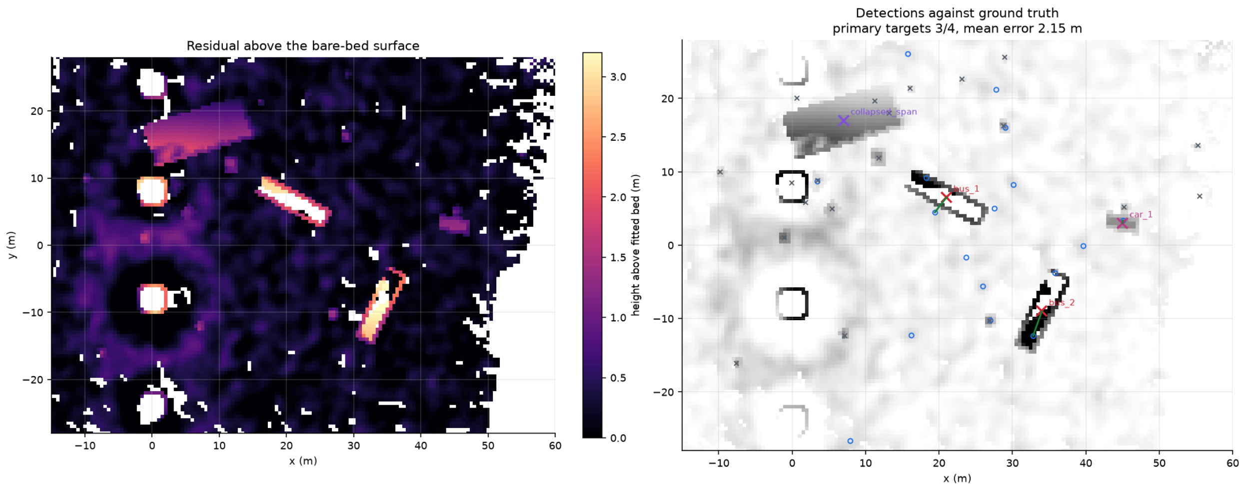


Figure 11: Left: height above the fitted bare bed surface. Right: detections against ground truth, with green lines joining each matched pair.

This number should be read with care. The dataset is a controlled water tank collection with a consistent background, so it is an easier problem than field sonar and near saturated performance is expected. It establishes that the perception pipeline is sound; it does not by itself predict field performance.

6.1 Domain transfer

Two transfer directions were examined. Running the real trained detector directly on simulated open river frames gives essentially no detections. The diagnosis is that the gap is dominated by scene content and capture geometry, not by acoustic physics: the tank captures are dense with walls, rig structure and cabling, at a short range window and a near grazing look, none of which resembles an open riverbed survey.

The operationally relevant direction is the reverse, and it is the one the system depends on, because there is no labelled sonar for Indian flood scenarios and there will not be one before deployment. Results for a detector trained only on simulator output and evaluated on the real held out test split are reported in Section 7.

7 Results

7.1 Reference mission

A single autonomous mission over the modelled Savitri site: deploy, acquire bottom lock, lawnmower search of the box, orbit inspection of two piers, return and report. No operator input after launch.

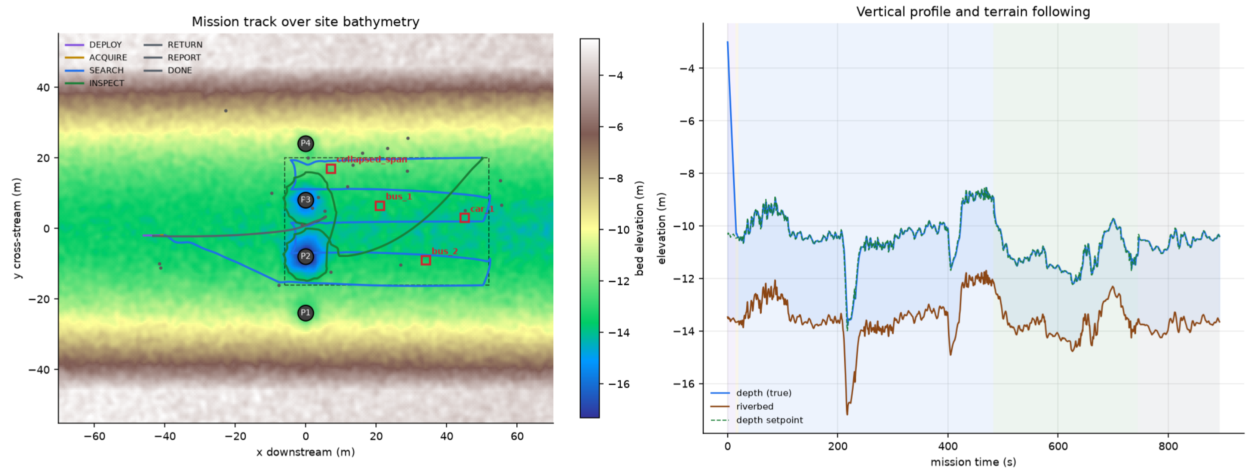


Figure 12: Mission track over site bathymetry, coloured by mission phase, and the vertical profile showing terrain following against the riverbed.

Table 6: Reference mission results. Navigation error is against simulation ground truth, which the vehicle never observes.

Metric	Value
Mission duration	893 s (14.9 min)
Path length	573 m
Sonar pings	1391
Soundings mapped	12,924,685
Search box coverage	95.3 %
DVL availability	96.4 %
Navigation error, mean	0.15 m
Navigation error, maximum	0.29 m
Navigation drift, fraction of distance travelled	0.04 %
Bathymetric map RMSE against truth	0.28 m

7.2 Scour measurement

Table 7: Scour measured autonomously against ground truth.

Pier	Measured depth	True depth	Recovery
P2	3.01 m	3.40 m	89 %
P3	1.76 m	2.10 m	84 %

Measured scour is systematically shallow by roughly 0.37 m. The cause is understood: the sonar footprint averages across the narrow bottom of the scour cone, the map cell is 0.5 m, and returns at very shallow grazing angles are rejected as poorly localised. The bias is consistent and therefore correctable by calibration, but it is reported here uncorrected.

7.3 Target detection, and what it costs

Both submerged vehicles are found in every run, at a mean localisation error of 1.80 m across the 5 seeds. The 2.15 m quoted in the table above is that same quantity for the single reference run tabulated there, which is

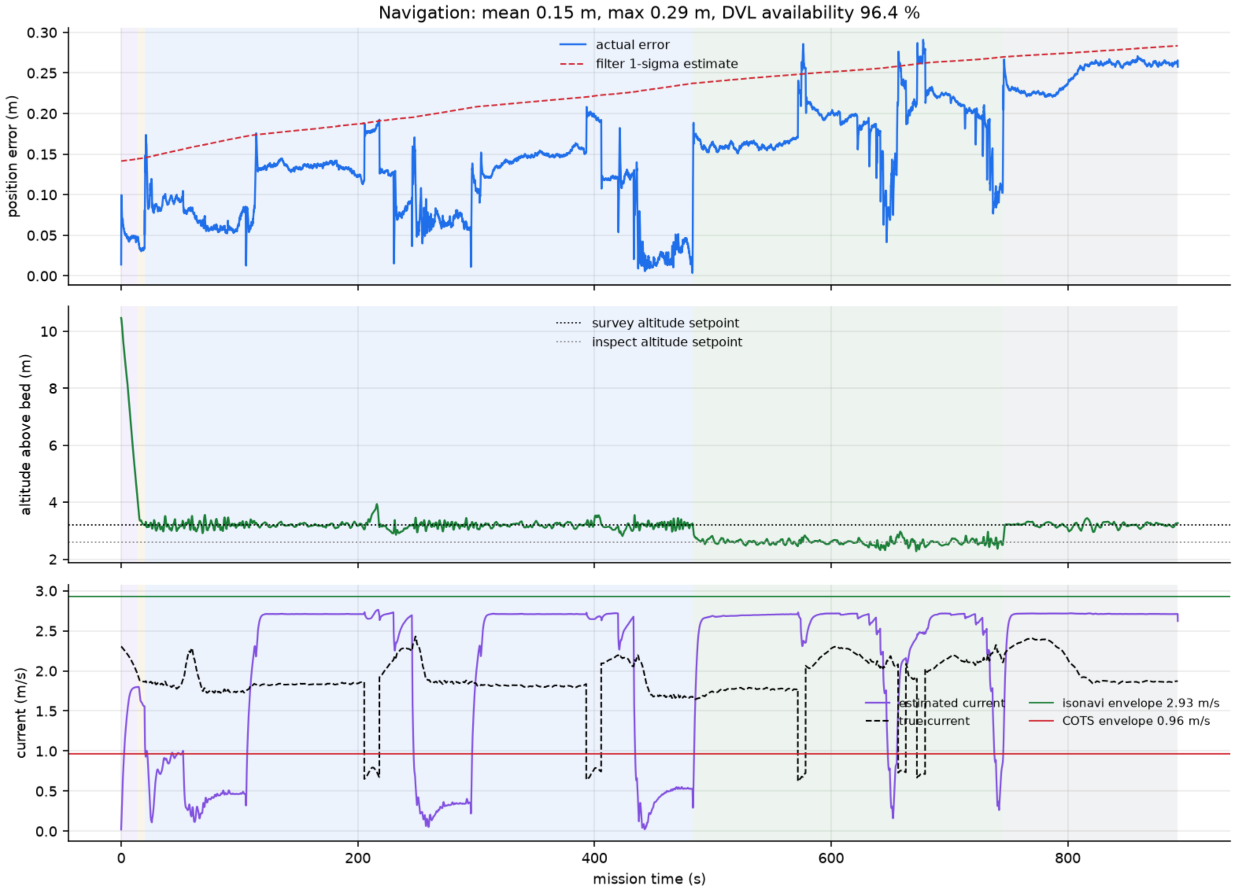


Figure 13: Navigation error against the filter's own uncertainty estimate, altitude tracking, and estimated against true current with both vehicle envelopes marked.

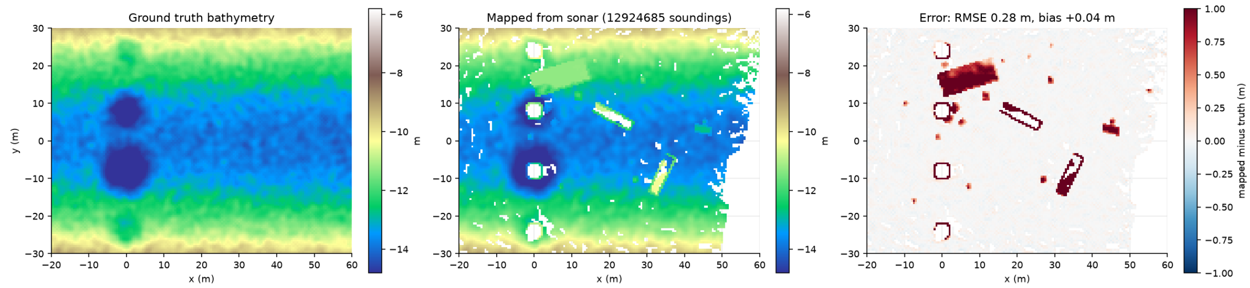


Figure 14: Ground truth bathymetry, the map built from simulated sonar during the mission, and the difference. The residual is near zero over open bed; the remaining structure is the submerged objects, which stand above the bare bed and are the subject of Section 7.

Table 8: Geometric detection from the bathymetric map, reference run. No training data is used. Small boulders fall below the map cell size and the sonar footprint and are not separable from bed texture.

Class	Detected	Present
Large vehicle (bus)	2	2
Small vehicle (car)	1	1
Structural debris	0	1
Boulders	3	14
Primary targets: 3/4, mean localisation error 2.15 m		

one of those seeds and sits inside the 0.40 m spread; the two numbers are consistent rather than competing. That is the operationally meaningful result for a search task: it places a diver or a recovery crane within a boat length of the target, in water where neither could otherwise be directed at all.

The collapsed deck span is the harder case and is missed in two of the 5 runs. It is a 15 m by 6 m slab, so it is matched footprint aware, against a gate of half its own diagonal rather than a fixed radius. In the runs that miss it the nearest contact still falls about a metre outside that footprint. An extended, low, flat object lying nearly flush with the bed is close to the worst case for a height threshold detector, and it is the case a trained classifier would most obviously help with.

7.3.1 The false alarm rate, which recall alone hides

Reporting recall by itself would flatter this. The detector flags any bed anomaly above its height threshold, so it also raises contacts that correspond to nothing in the scene. Matching every contact against every registered object, boulders and debris included, separates the two:

Table 9: Detection quality over 5 runs on a 1.54 ha survey box.

Contacts raised per run	24.4
Of which false alarms	15.4 ± 1.5
False alarms per hectare	10.0
All primary targets found	3/5 runs

Roughly 15.4 of the 24.4 contacts are spurious, and their median distance to the nearest real object is about 8.6 m, so they are not near misses on real boulders but genuine false returns from map noise, mostly near the survey boundary where sounding density is lowest.

Stated plainly, this is a screening tool and not a classifier. What it delivers is the reduction of an unsearchable 1.54 ha of zero visibility water to roughly 24.4 georeferenced contacts, with the submerged vehicles reliably among them. For a dive team with hours of bottom time that is the difference between a search and a task list. It is not a finished product, and the obvious next step is a classifier over the contact list rather than a better threshold.

7.4 Cross-domain transfer: a negative result, stated plainly

A detector was trained on 2200 simulated frames with analytically derived labels and no human annotation, then evaluated on the real ARIS held out test split. The reverse direction was measured the same way, so the two are directly comparable: standard detection mAP on a held-out split of the other domain.

Table 10: Symmetric cross-domain evaluation. Both models are strong inside their own domain and both fail across the gap.

Trained on	Tested on	mAP@0.5	mAP@0.5:0.95
Real sonar	real test split	99.0 %	78.1 %
Simulator	simulated val split	84.6 %	61.1 %
Real sonar	simulated val split	0.8 %	0.1 %
Simulator	real test split	0.4 %	0.1 %

The two diagonal entries matter as much as the off-diagonal ones. Each model is strong on its own data, so neither the simulated imagery nor its labels are defective: the simulated frames are internally consistent and genuinely learnable. The gap is between the domains, and it is close to symmetric, which is itself informative. A one-sided failure would suggest the simulator were simply too easy or too clean; a symmetric one says the two distributions are far apart in a way that neither model bridges.

The diagnosis is that the residual gap is dominated by scene content and capture configuration rather than by the acoustic physics. The public data is a water tank dense with wall returns, mounting rig and cabling, imaged at a short range window and a near grazing look, and its classes are centimetre scale objects whose signature depends on fine structure well below what a geometric model represents. None of that resembles an open riverbed survey of metre scale targets.

Two conclusions follow, and both shape the system rather than being excuses. First, the simulator should not be presented as a drop in replacement for real training data for fine grained classification. Second, and more usefully, the capabilities this system actually depends on do not require that transfer at all. Locating a submerged bus is a geometry problem, solved by the detector of Section 7 at 3/4 with no training data whatsoever. Where a learned classifier is needed, the relevant question is not zero-shot transfer but whether simulator pretraining reduces how much real data must be collected, which is measured next.

7.5 Data efficiency: simulator pretraining does not help either

If zero-shot transfer fails, the next question is whether simulator pretraining at least reduces how much real data must be collected. This is the question that would decide whether the simulator earns its place in the perception pipeline, so it was measured directly. Two arms, identical except for initialisation, were trained on the same N real images and evaluated on the same untouched real test split: one starting from generic natural-image weights, one from the simulator-trained weights.

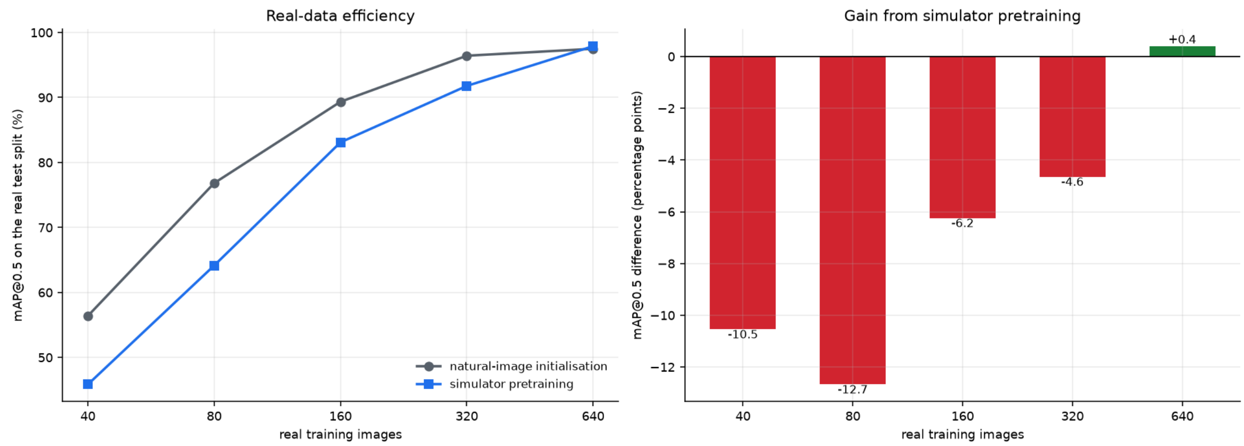


Figure 15: Real-data efficiency. Both arms are trained on the same real images and evaluated on the same real test split, differing only in initialisation. Simulator pretraining is worse at every size except the largest.

Table 11: Detection mAP@0.5 on the real test split against the number of real training images.

Real images	Natural-image init	Simulator init	Difference
40	56.4 %	45.8 %	−10.5
80	76.8 %	64.1 %	−12.7
160	89.3 %	83.1 %	−6.2
320	96.4 %	91.8 %	−4.6
640	97.5 %	97.9 %	+0.4

The result is negative and is reported as such. Simulator pretraining is worse than generic pretraining at every data size that matters, and only draws level once enough real data is present that the initialisation no longer matters. The likely mechanism is ordinary negative transfer: natural-image pretraining supplies broad low-level features, whereas pretraining on a narrow synthetic sonar distribution overwrites them with domain-specific features that do not match the real sensor.

This settles the question honestly. The simulator is not a useful training source for this classification problem, and no amount of presentation should suggest otherwise.

What the simulator is genuinely for is threefold, and none of it depends on this transfer:

1. **Closed-loop autonomy validation.** Every navigation, control and mission result in Section 7 exists because the vehicle can be flown against physically modelled sensors and hydrodynamics.
2. **Design decisions that cannot be made any other way.** The finding that the standard commercial platform is out of envelope, and the fin requirement that follows from the Munk moment, came from the simulation and changed the vehicle.
3. **The geometric detector.** Locating a submerged bus is a geometry problem, and both are found in every run with no training data at all. That is the capability the disaster mission actually needs, and it is precisely the one for which no labelled sonar exists. It is a screening tool rather than a classifier: it also raises about 15.4 false contacts per run, and it misses the flat deck slab in two runs of 5.

7.6 Repeatability

A single run is an anecdote. The mission was repeated across 5 independent seeds, re-randomising sensor noise, DVL dropout pattern and sonar speckle while holding the scene fixed.

Table 12: Mission repeatability across 5 independent runs, mean and standard deviation.

Metric	Mean ± SD
Search-box coverage	95.3 ± 0.2 %
Navigation error, mean	0.322 ± 0.161 m
Navigation error, maximum	0.663 ± 0.293 m
DVL availability	96.0 ± 0.3 %
Bathymetric map RMSE	0.285 ± 0.001 m
Target localisation error	1.80 ± 0.40 m
Scour depth recovery	86.1 ± 0.1 %
Missions completed	5/5
All primary targets found	3/5

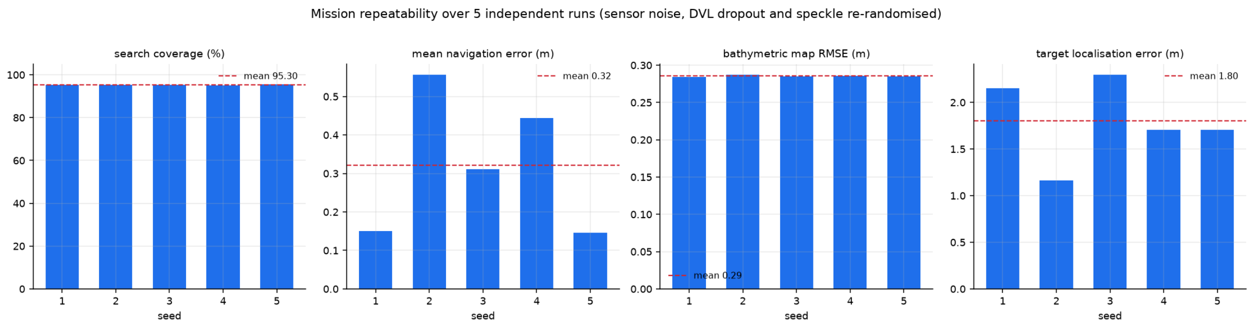


Figure 16: Per-seed results with the mean marked. The spread, not the best run, is the honest statement of what the system does.

7.7 Endurance, which the current sets

A hovering vehicle answers the endurance question worst, because holding station in a current costs full thrust while doing no useful work at all. It is therefore worth computing rather than quoting a transit figure.

Propeller power is taken from momentum theory, $P = T^{3/2}/\sqrt{2\rho A}$, divided by a drive efficiency of 0.60 covering motor, controller and propeller. That relation is superlinear in thrust, so the fleet total cannot be recovered from the peak thruster; the mission log records all eight forces and the integral runs over them.

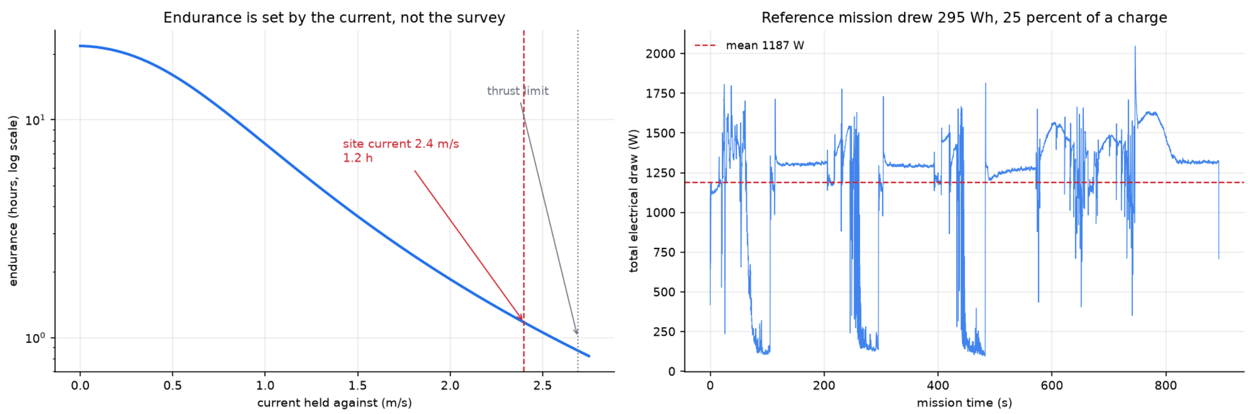


Figure 17: Left: endurance holding station, against the current being fought. Right: the electrical draw of the reference mission, integrated from the logged thruster forces. The low sections are downstream legs, the high ones are upstream legs and station keeping.

The reference mission drew 295 W h, which is 25 % of a usable charge, so the vehicle carries 4.1 such missions between charges. Mean draw was 1187 W against a hotel load of only 55 W: propulsion is essentially the entire energy budget.

The endurance curve is the honest part. In slack water the same battery lasts 17.8 h. Holding against the design current of 2.4 m s^{-1} it lasts 1.2 h. That is a factor of 15 across the operating range, and it is a direct consequence of the hover authority that makes the vehicle useful here at all.

Two things follow for operations. Endurance must be quoted against a current, never as a single number, and the mission plan has to work the flow rather than ignore it, which is why survey legs run along the flow axis and not across it.

8 Hardware in the loop

Everything above is simulation. The maturity question a funder actually asks is whether the autonomy runs on real hardware, so it was put on real hardware. The control loop was distributed across three processors:

- a host, running the physics, the sonar and the sensor models, holding the ground truth;
- a flight computer, a LicheeRV Nano single-board computer with a RISC-V processor at 750 MHz, running the estimation, control and mission state machine;
- an actuator interface, an ESP32 generating the eight electronic speed controller channels.

The board is the real-time master. Each tick it requests the sensor sample from the host, runs the extended Kalman filter and the controller, sends pulse widths to the ESP32, and returns the applied wrench so the host integrates the plant. The estimation, control and mission code on the board are the exact same modules the pure-simulation stack runs, not a reimplementation.

The board never receives ground truth. It sees only what the sensors produce and acts on its own estimate, so a matching mission result means the autonomy runs on the hardware, not that the hardware was handed the answer. The link is a framed, CRC-checked protocol that resynchronises after corruption and rejects bad frames rather than acting on them; it passed the full mission with zero CRC errors.

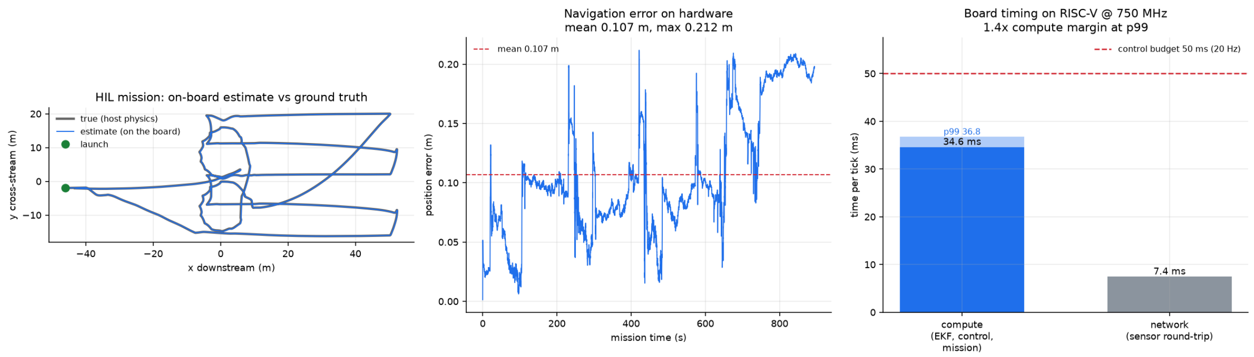


Figure 18: Full mission executed on the flight computer. Left: the on-board pose estimate against the host's ground truth. Centre: navigation error over the run. Right: per-tick board timing, compute separated from the sensor round-trip, against the control budget.

Table 13: The identical mission, run in pure simulation and with the control loop executing on the RISC-V flight computer.

	Pure simulation	On the flight computer
Navigation error, mean	0.15 m	0.107 m
Navigation error, maximum	0.29 m	0.212 m
Ticks completed	17 834	17,884
Final phase	DONE	DONE
CRC errors	n/a	0

The mission completed on the board with a mean navigation error of 0.107 m, within centimetres of the pure-simulation result. The estimation, control and mission logic cost 34.6 ms per tick on the board, 36.8 ms at the 99th percentile, against a 50 ms control budget at 20 Hz, a margin of 1.4 times. The sensor round-trip added a further 7.4 ms, which is an artifact of the bench topology where the plant runs on a separate host; on the vehicle the estimator and the plant sensors are on the same board.

8.1 The interface that turns that allocation into pulses

An allocation that never reaches an actuator is an opinion. The eight thruster forces the controller produces are therefore converted to pulse widths on the flight computer and sent, over the same framed and CRC checked protocol, to an ESP32 that generates eight real hardware channels at the standard 50 Hz speed controller frame.

The value it sends back is read from the timer's duty registers rather than from the variable it was given, so the echo reports what the hardware is emitting. Bench measurement puts the agreement at zero microseconds across the full 1000 µs to 2000 µs range, at a resolution of 0.305 µs per bit.

Table 14: Actuator interface over the complete mission.

Pulse width frames commanded	17,884
Echoes returned	17,879
Echoes exactly reproducing a commanded width	17,878
Frames not reproduced	1
Pipeline lag	2.3 frames
Cost to the control loop	3.3 ms mean, 3.8 ms at the 99th percentile
CRC errors	0

One frame in nearly eighteen thousand was not reproduced. The interface costs the control loop 3.3 ms per tick, which takes the loop from 37 ms to 42 ms against its 50 ms budget, with the 99th percentile at 49.3 ms. That is a real cost and it is measured rather than argued: the same mission was run with the interface disconnected to obtain the comparison.

Two honest qualifications. The echo answers a command roughly 2.3 frames back, because the interface reports the registers as it finds them and then applies the new command, which avoids blocking a whole 20 ms frame to confirm each one. And on this bench the link is a socket through a tunnel and a serial bridge, because the microcontroller enumerates on a different machine from the flight computer; on the vehicle it is a direct connection between two boards, and the few milliseconds of transport in the figure above would not be there.

What matters is what the vehicle would do with a thruster attached to any of those eight channels: the pulse train is being generated, from the autonomy's own output, on the target hardware, for the length of a full mission.

8.1.1 The same trajectory on two instruction sets

The mission was also run with both halves of the loop on the host, over a loopback socket, so that the only difference from the hardware run is which processor executed the autonomy. The two agree more closely than expected: the host returns a mean navigation error of 0.106 865 873 713 and the RISC-V board returns 0.106 865 873 714, over 17,884 ticks.

Agreement to roughly one part in 10^{12} is the signature of the same arithmetic being performed in the same order, not of two implementations that happen to behave alike. The estimator, the controller and the mission logic are therefore genuinely portable across the two architectures rather than merely tuned to produce similar behaviour on each, and a discrepancy in a future run would point at the port rather than at the physics.

This is the difference between a claim of simulated capability and a claim of validated capability. The autonomy is real code running inside a real-time budget on representative flight hardware, and it reproduces the simulation.

9 Technology readiness, stated per subsystem

Readiness is not a property of a project, it is a property of each thing in it, and quoting one number for a system whose parts sit at different levels hides more than it says. The standard definition of level 4 is component or breadboard validation in a laboratory environment, with components integrated to establish that they work together. Against that:

The position is precise: the autonomy and its actuation path are validated on hardware, the perception is validated on real data, and the vehicle is validated on paper. That is a deliberate ordering rather than an

Table 15: Readiness by subsystem, with the evidence each rests on.

Subsystem	Level	Evidence
Autonomy: estimation, control, mission logic	4	Executes on a RISC-V flight computer inside a 50 ms real-time budget, driving a separate microcontroller over a CRC checked link, for a complete 894 s mission. Reproduces the simulated trajectory to 10^{-12} across two instruction sets.
Actuator interface	4	Eight hardware PWM channels generated on the target microcontroller from the allocation's own output, measured back from the duty registers, exact across the full 1000 µs to 2000 µs range, and 17,878 of 17,879 echoes exact over a complete mission.
Acoustic perception	4	Trained and evaluated on real sonar, not simulation: 99.0 % on a held out real test split.
Geometric detection and mapping	3	Validated against simulated ground truth only. No real bathymetry has been mapped by this vehicle.
Vehicle structure and hydrodynamics	3	Complete parametric CAD with a closed mass, buoyancy and trim budget, closed form depth and drag analysis, cross checked against a geometry kernel. No physical article exists and nothing has been in water.

accident of what was easy. The autonomy is the part that is difficult to buy, difficult to copy and specific to this problem; hulls, thrusters and pressure housings are none of those things, and Section 11 shows they are also the cheap part of the vehicle.

What this programme needs next is therefore not more analysis. It is water: a tank, then a reservoir, then a river, with the instrumented vehicle the analysis already specifies. That is precisely the transition from level 4 to levels 5 and 6, and it is the step where testing facilities and an operational partner change the timeline more than funding alone would.

10 Verification and limits

Every quantitative claim in this report is produced by a script in the repository and regenerated from the measured result files, so the document cannot drift out of step with the experiments.

10.1 What is verified

- Ray primitive intersection against analytic values to 10^{-9}.
- Sonar shadow and target contrast against physical expectation.
- Vehicle buoyancy, drift under no thrust, and thrust envelope against closed form drag balance.
- Navigation error against simulation ground truth over a full mission.
- Mapped bathymetry, scour and target positions against ground truth.
- Perception on a real, held out sonar test split.
- The full mission executing on a RISC-V flight computer, reproducing the simulation result within centimetres inside the control budget.
- A closed mass and stability budget: the modelled hull encloses the simulated displacement to 0.001 %, at the simulated mass, in level trim.
- The analytic layout against an independent geometry kernel: hull volume, volume centroid, mass, displacement and trim all agree, which is what caught both the stability and the trim errors.

10.2 What remains to be measured

- **No physical vehicle exists.** All vehicle results are simulation. The hydrodynamic coefficients are now cross checked against estimates derived from the CAD geometry, and agree to 21 % in axial drag, but both sides of that comparison are closed form. CFD and tow tank validation remain required, and the appendage drag coefficients are the least certain part.

- **The site is representative, not surveyed.** Dimensions are consistent with published descriptions of the Mahad collapse and with IRC:SP:35 practice, but they are modelling assumptions, not measured bathymetry.
- **The sediment attenuation term is an approximation**, not a first principles result.
- **Zero-shot transfer from tank data to open river frames fails.** This is reported rather than hidden, with the diagnosis above.
- **Detection recall is quoted for objects above roughly one metre.** Smaller debris is below the resolution of the map cell and the sonar footprint.
- **The detector's false alarm rate is high.** About 10.0 spurious contacts per hectare accompany the real ones, and the flat deck slab is missed in two runs of 5. The output is a contact list for a dive team, not an identification.
- **Not single fault tolerant for station keeping.** Losing one of the four horizontal thrusters halves surge and drops the holdable current to $1.86\,\mathrm{m\,s^{-1}}$, below the design condition. Attitude and depth control survive any single failure.
- **The stability margin is now CAD derived, not assumed**, but the hydrodynamic coefficients around it remain estimates. The 27.6 mm separation replaced an assumed 85 mm that the layout showed to be unreachable.

11 What it costs, and where it sits

11.1 Bill of materials

One airframe, prototype quantity. This is materials and bought-in equipment only. It excludes design labour, integration, test, tooling and qualification, which for a first article normally exceed the parts, so it is not a unit price and is not offered as one.

Table 16: Bill of materials by subsystem, one airframe. The doppler log is a published price; the imaging sonar is quotation only and is an estimate carrying real uncertainty.

Subsystem	USD	Share
Acoustics, sonar and doppler log	23 710	72 %
Structure, hull and fittings	5250	16 %
Propulsion, eight thrusters and drives	1880	6 %
Avionics, computing and inertial	1135	4 %
Power	850	3 %
Total	32,825	
27.6 lakh INR at 84 per USD		

The distribution is the interesting part. The two acoustic instruments are 72 % of the bill. The hull, the propulsion, the power system and the entire autonomy stack together come to 9115 USD.

That has a direct consequence for where effort is worth spending. The work in this report, the estimation, the control, the mission logic and the geometric detector, runs on a computer that costs a few hundred dollars, and it is the part that is genuinely ours. The cost sits in two imported instruments. If this programme is to reduce cost further, the lever is the sonar, not the vehicle, and that is a separate and much larger undertaking than the one proposed here.

11.2 Where this vehicle sits

Both comparators are mature, capable vehicles and neither is being criticised. They are transit survey torpedoes: a single propeller with control fins, which requires forward speed for control authority and provides no lateral or vertical thrust. That is the right design for running long survey lines in open water, and it is why they are faster, deeper rated and longer legged than this vehicle.

It is also why neither can do the job described in Section 7. Holding station over a scoured pier in a $2.4\,\mathrm{m\,s^{-1}}$ current, and orbiting it to measure the hole, requires the vehicle to generate force in every direction at zero forward speed. isonavi-1 gives up depth, endurance and half its drag budget to buy that, which is a poor trade for a survey vehicle and the only workable one for a disaster vehicle.

Table 17: Comparable compact survey vehicles. The price column mixes order price against a parts bill and is a scale reference, not a like for like comparison.

	REMUS 100	Iver3	isonavi-1
Length	1.6 m	2.1 m	1.07 m
Diameter	0.19 m	0.14 m	0.18 m
Mass	under 45 kg	about 30 kg	28.0 kg
Depth rating	100 m	100 m	50 m
Propulsion	prop and fins	prop and fins	eight vectored
Hover and station keep	no	no	to 2.69 $\mathrm{m\,s^{-1}}$
Designed for	open water survey	open water survey	flood debris
Cost, USD	order 300 000	order 150 000	parts 32,825

The claim is therefore not that this is a better AUV. It is that it is a different one, aimed at a mission the available platforms are not built to attempt.

12 Development pathway

The present state is a verified simulation and autonomy stack with perception validated on real sonar data. The path to a field system is:

Table 18: Development pathway.

Phase	Work	Evidence produced
Now	Simulation, autonomy, mapping and perception; a closed design loop from the vehicle CAD back into the physics; the full mission executing on a RISC-V flight computer	This report and the repository; a mass, stability and energy budget that closes; the autonomy validated on flight hardware
Months 1 to 6	CFD and tow tank validation of the faired hull; a pressure test of the hull section; indigenous thruster and pressure housing sourcing	Measured hydrodynamic coefficients in place of the closed form estimates here; a qualified depth rating
Months 7 to 12	Integrated vehicle; tank and reservoir trials; DVL and sonar integration	Wet testing against surveyed targets
Months 13 to 24	Instrumented river trials with a partner agency; field validation of scour measurement against diver survey	Field data at operational current and turbidity

The autonomy and perception software is the defensible indigenous content, and the bill of materials in Section 11 shows why that is the right place to have spent the effort: it runs on hardware costing a few hundred dollars, while 72 % of the vehicle cost sits in two imported acoustic instruments.

The hardware baseline therefore starts from proven components to remove mechanical risk, and is intended to be progressively indigenised in cost order rather than in order of ease. The imaging sonar is the largest single line and is the eventual target; the doppler log is second; the pressure housing and thrusters are the near term ones because they are the achievable ones. Nothing in this report depends on that sequence being completed, which is the point of having measured where the money actually goes.

Reproducing these results

```
python tests/smoke_sonar.py     sonar physics verification
python tests/smoke_control.py   dynamics, estimation, control verification
python eval/thrust_envelope.py  vehicle envelope analysis
python eval/run_mission.py      full autonomous mission
python eval/make_figures.py     report figures
python eval/detect_eval.py      geometric target detection
python eval/sim2real.py         zero-shot transfer test

cad/isonavi_layout.py           mass, buoyancy and trim budget
cad/isonavi_cad.py              geometry, STEP and STL
cad/verify_cad.py               layout against the geometry kernel
cad/structures.py               depth rating and pylon loads
cad/hydrodynamics.py            drag and added mass from the geometry
cad/energy.py                   power and endurance
cad/redundancy.py               thruster-out envelope
cad/bom.py                      bill of materials

hil/test_hil_loopback.py        full loop, no hardware
hil/deploy_board.py             push the autonomy to the flight computer
hil/run_hil.py                  mission on the real board
```